



Document



Vision Encoder



$O(N^2)$



□ Background Token ■ Text Token



4x Projector



LLM Decoder



result



Document



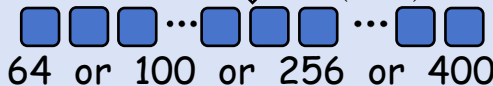
DeepEncoder



Window Attention
Convolution Compressor



$O(Nw)$



Projector



LLM Decoder



result



Document



Valid Region Focus Module
(VRFM)



Vision Encoder



$O(km_i^2)$



4x Projector



LLM Decoder



result

Ours

Document



Vision Encoder Normal



$O(N^2)$



Vision Encoder BlackHole



$O(m^2)$



1x Projector



LLM Decoder



result